



National Textile University
Department of Computer Science

Subject

Operating System

Submitted to:

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Submitted by:

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Registration Number

23-NTU-CS-1128

Home Task

01

Semester

5th

Verification of WSL and Ubuntu installation:

```
Terminal Close Tab
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26100.6584]
(c) Microsoft Corporation. All rights reserved.

C:\Users\pc>wsl -l -v
NAME STATE VERSION
* Ubuntu-24.04 Stopped 2

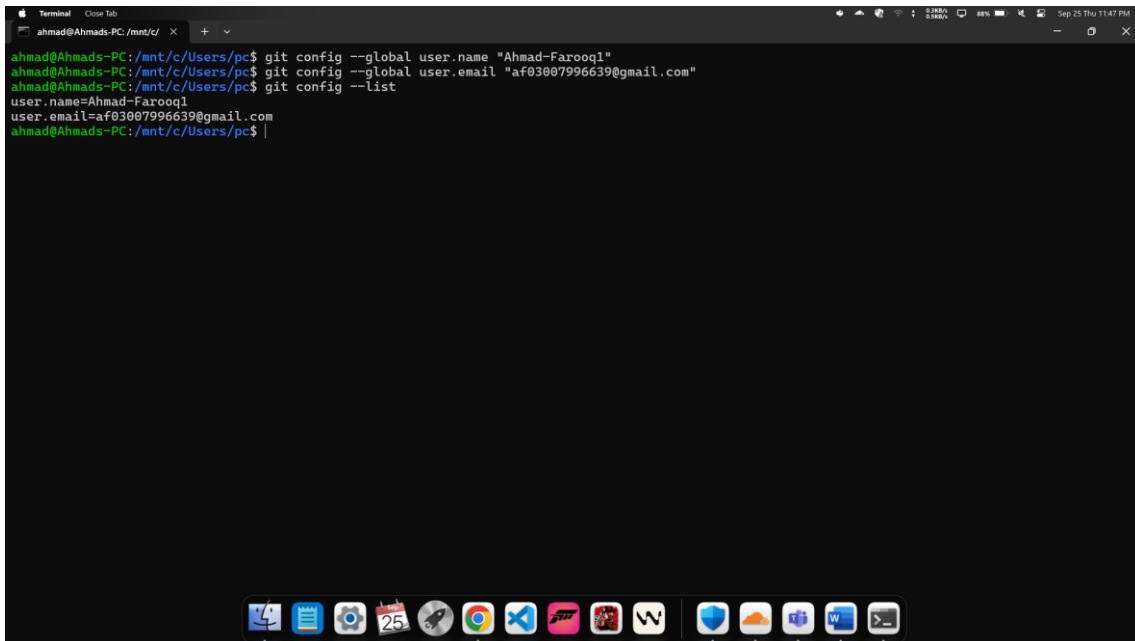
C:\Users\pc>
```

Update Ubuntu Environment:

```
Terminal Close Tab
ahmad@Ahmads-PC: /mnt/c/ x + v
ahmad@Ahmads-PC:/mnt/c/Users/pc$ sudo apt update && sudo apt upgrade -y
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1171 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1443 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [282 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [15.3 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1485 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [198 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [8744 B]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [880 kB]
54% [14 Components-amd64 134 kB/378 kB 35%] [15 Packages 4717 B/880 kB 1%] 443 kB/s 13s
```

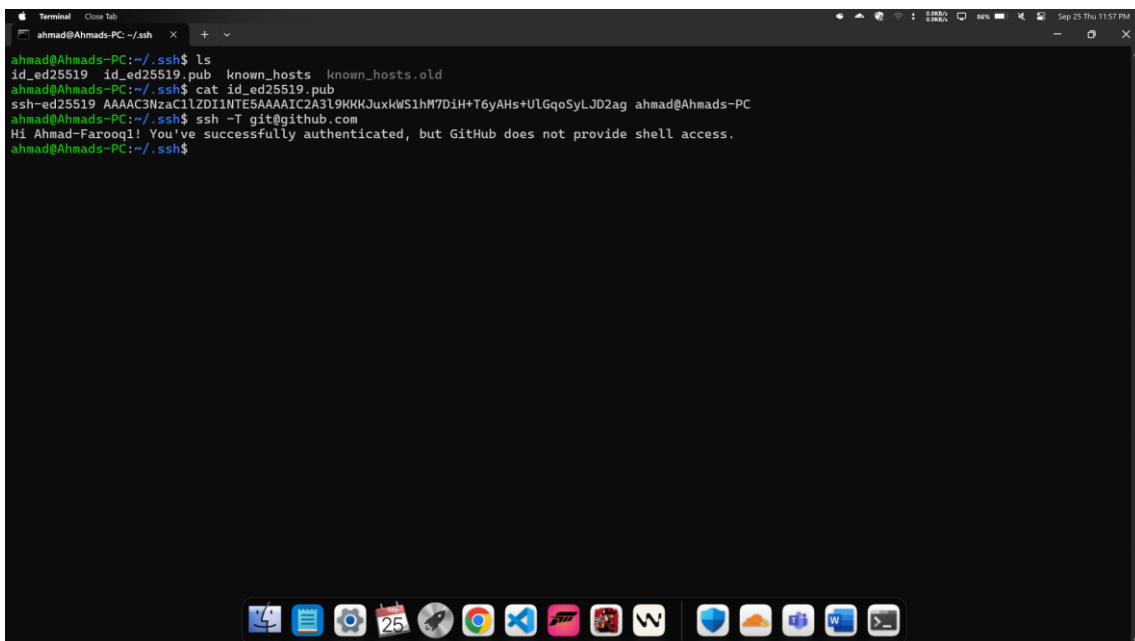
Git & GitHub

UserName and Email configuration:



```
ahmad@Ahmads-PC: /mnt/c/Users/pc$ git config --global user.name "Ahmad-Farooq1"
ahmad@Ahmads-PC: /mnt/c/Users/pc$ git config --global user.email "af03007996639@gmail.com"
ahmad@Ahmads-PC: /mnt/c/Users/pc$ git config --list
user.name=Ahmad-Farooq1
user.email=af03007996639@gmail.com
ahmad@Ahmads-PC: /mnt/c/Users/pc$
```

Testing SSH key:



```
ahmad@Ahmads-PC: ~/ssh$ ls
id_ed25519 id_ed25519.pub known_hosts known_hosts.old
ahmad@Ahmads-PC: ~/ssh$ cat id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1IOTE5AAAIC2A3l9KKKJuxkWS1hM7D1H+T6yAHs+ULGqoSylJD2ag ahmad@Ahmads-PC
ahmad@Ahmads-PC: ~/ssh$ ssh -T git@github.com
Hi Ahmad-Farooq1! You've successfully authenticated, but GitHub does not provide shell access.
ahmad@Ahmads-PC: ~/ssh$
```

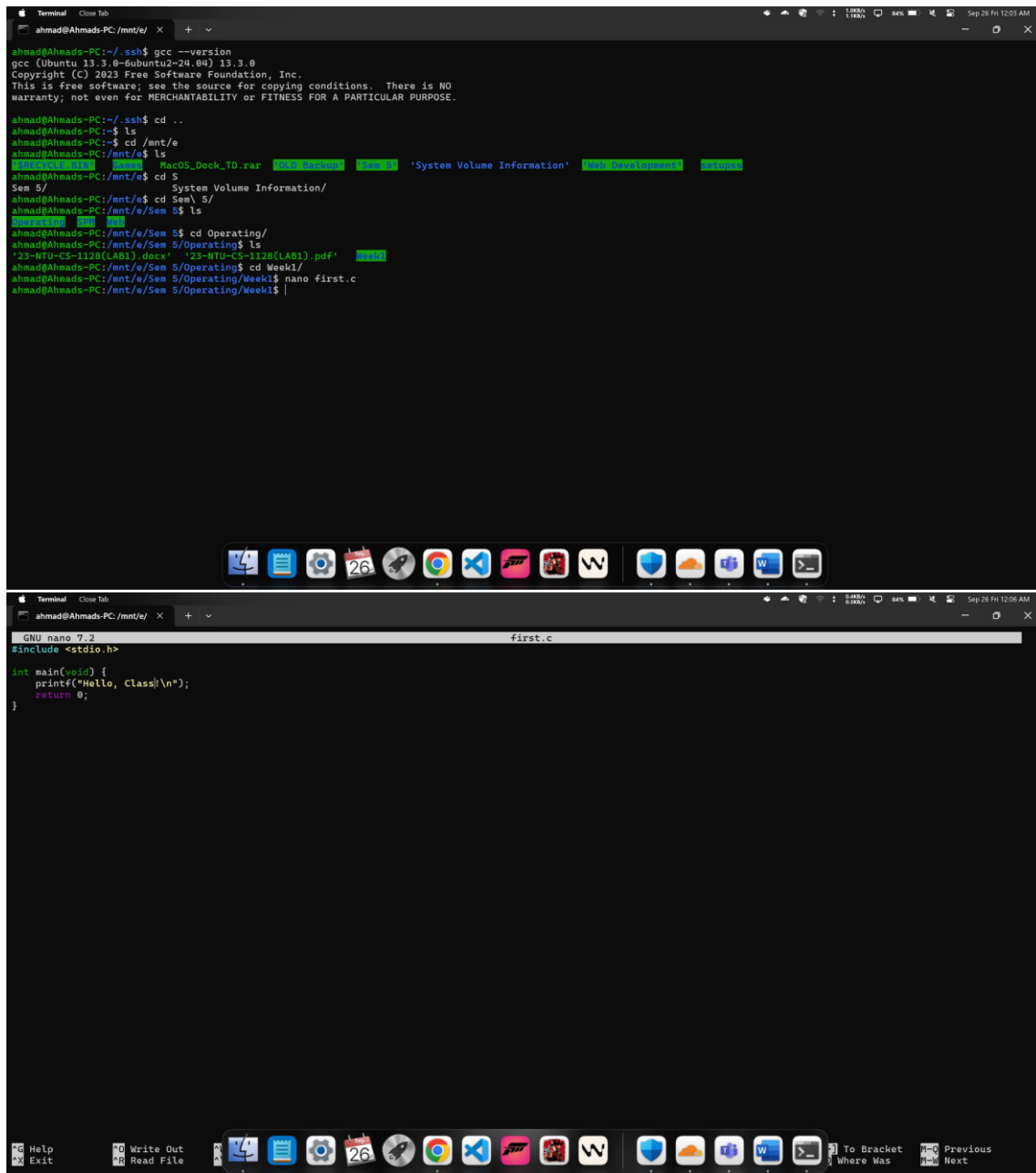
Install Build Tools:

```
Terminal Close Tab
ahmad@Ahmads-PC: ~/.ssh x + v
ahmad@Ahmads-PC:~/.ssh$ sudo apt install build-essentials
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package build-essentials
ahmad@Ahmads-PC:~/.ssh$ sudo apt install build-essential
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  bzip2 dpkg-dev fakeroot g++ g++-13 g++-13-x86-64-linux-gnu g++-x86-64-linux-gnu libalgorithm-diff-perl libalgorithm-diff-xs-perl
  libalgorithm-merge-perl libdpkg-perl libfakeroot libfile-fcntllock-perl libstdc++-13-dev lto-disabled-list make
Suggested packages:
  bzip2-doc debconf-keyring g++-multilib g++-13-multilib gcc-13-doc bsr libstdc++-13-doc make-doc
The following NEW packages will be installed:
  build-essential bzip2 dpkg-dev fakeroot g++ g++-13 g++-13-x86-64-linux-gnu g++-x86-64-linux-gnu libalgorithm-diff-perl
  libalgorithm-diff-xs-perl libalgorithm-merge-perl libdpkg-perl libfakeroot libfile-fcntllock-perl libstdc++-13-dev lto-disabled-list make
0 upgraded, 17 newly installed, 0 to remove and 0 not upgraded.
Need to get 16.4 MB of archives.
After this operation, 61.3 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libstdc++-13-dev amd64 13.3.0-6ubuntu2-24.04 [2420 kB]
0% [1 libstdc++-13-dev 47.5 kB/2420 kB 2%]
```

GCC version :

```
Terminal Close Tab
ahmad@Ahmads-PC: ~/.ssh x + v
ahmad@Ahmads-PC:~/.ssh$ gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2-24.04) 13.3.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
ahmad@Ahmads-PC:~/.ssh$ |
```

Creating First C file:



The image consists of two screenshots of a macOS terminal window, showing the steps to create a C file named 'first.c'.

Top Screenshot: The terminal shows the user navigating through the file system. The prompt is `ahmad@Ahmads-PC: /mnt/e/`. The user runs `gcc --version`, which outputs the GCC version 13.3.0. Then, the user runs `cd ..`, `ls`, `cd /mnt/e`, and `ls`. The output of `ls` shows a directory named `MacOS_Dock_TD.rar` and a file named `'System Volume Information'`. The user then runs `cd S`, which changes the directory to `Sem 5/`. The user runs `cd Sem\ 5/`, and the prompt changes to `ahmad@Ahmads-PC: /mnt/e/Sem 5/`. The user runs `ls`, and the output shows a directory named `Operating/`. The user runs `cd Operating/`, and the prompt changes to `ahmad@Ahmads-PC: /mnt/e/Sem 5/Operating/`. The user runs `ls`, and the output shows a file named `'23-WTU-CS-1128(LAB1).pdf'`. The user runs `cd Week1/`, and the prompt changes to `ahmad@Ahmads-PC: /mnt/e/Sem 5/Operating/Week1/`. The user runs `nano first.c`, and the prompt changes to `ahmad@Ahmads-PC: /mnt/e/Sem 5/Operating/Week1/`.

Bottom Screenshot: The terminal shows the user editing the file `first.c` using the `nano` editor. The prompt is `GNU nano 7.2`. The user has entered the following code:

```
#include <stdio.h>

int main(void) {
    printf("Hello, Class!\n");
    return 0;
}
```

The bottom screenshot also shows the `nano` editor's status bar at the bottom, which includes options like `Help`, `Exit`, `Write Out`, `Read File`, `To Bracket`, and `Previous`.

Executing C file:

```
Terminal Close Tab
ahmad@Ahmads-PC: /mnt/e/

ahmad@Ahmads-PC:~/$ gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

ahmad@Ahmads-PC:~/$ ssh$ cd ...
ahmad@Ahmads-PC:~/$ ls
ahmad@Ahmads-PC:~/$ cd /mnt/e
ahmad@Ahmads-PC:/mnt/e$ ls
MacOS_Dock_TD.rar 'System Volume Information'
ahmad@Ahmads-PC:/mnt/e$ cd S
Sem 5/
ahmad@Ahmads-PC:/mnt/e$ cd Sem\ 5/
ahmad@Ahmads-PC:/mnt/e/Sem 5$ ls
Operating/
ahmad@Ahmads-PC:/mnt/e/Sem 5$ cd Operating/
ahmad@Ahmads-PC:/mnt/e/Sem 5/Operating$ ls
'23-NTU-CS-1128(LAB1).docx' '23-NTU-CS-1128(LAB1).pdf'
ahmad@Ahmads-PC:/mnt/e/Sem 5/Operating$ cd Week1/
ahmad@Ahmads-PC:/mnt/e/Sem 5/Operating/Week1$ nano first.c
ahmad@Ahmads-PC:/mnt/e/Sem 5/Operating/Week1$ gcc first.c -o first.out
ahmad@Ahmads-PC:/mnt/e/Sem 5/Operating/Week1$ ./first.out
Hello, Class!
ahmad@Ahmads-PC:/mnt/e/Sem 5/Operating/Week1$
```

Pushing to Github:

```
Terminal Close Tab
ahmad@Ahmads-PC: /mnt/e/

git init
git add README.md
git commit -m "Pushing Assignment"
git branch -m main
git remote add origin git@github.com:Ahmad-Farooqi/Operating-System-.git
git push -u origin main
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /mnt/e/Sem 5/Operating/..git/
[master (root-commit) 69db9df] Pushing Assignment
1 file changed, 1 insertion(+)
create mode 100644 README.md
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 244 bytes | 2.00 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:Ahmad-Farooqi/Operating-System-.git
 * [new branch]    main -> main
branch 'main' set up to track 'origin/main'.
ahmad@Ahmads-PC:/mnt/e/Sem 5/Operating$ git add .
ahmad@Ahmads-PC:/mnt/e/Sem 5/Operating$ git commit -m "adding all files"
[main 7148edd] adding all files
6 files changed, 6 insertions(+)
create mode 100644 23-NTU-CS-1128(LAB1).docx
create mode 100644 23-NTU-CS-1128(LAB1).pdf
create mode 100644 Week1/23-NTU-CS-1128(Week1 HomeTask).docx
create mode 100644 Week1/first.c
create mode 100644 Week1/first.out
create mode 100644 Week1/~WRL0005.tmp
ahmad@Ahmads-PC:/mnt/e/Sem 5/Operating$ git push origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 12 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (8/8), 3.26 MiB | 1.04 MiB/s, done.
Total 8 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:Ahmad-Farooqi/Operati:
 69db9df..7148edd main -> main
ahmad@Ahmads-PC:/mnt/e/Sem 5/Operat:
```